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Q1) **6 marks**

Which one of the following code generates a set of numbers containing the number **12**?

A) `range(12)`
B) `range(0, 12, -1)`
C) `range(2, 12)`
D) `range(12, 0, -5)`
E) `range(1, 24, 2)`

Answer (A/B/C/D/E): _____

Q2) **8 marks**

Here is a description of the Hong Kong 'Mark 6' lottery:

> *"Mark Six is a lottery style game where players guess what numbers will be drawn out of a lottery machine. The lottery machine is a transparent plastic cylinder which contains colored balls numbered 1 to 49. The cylinder is rotated to provide randomization in the drawing process. Players choose six different numbers out of a possible 49."*

Somebody has created a Python program to generate a set of 6 numbers which can be used for the lottery. The program generates 6 unique numbers out of the 49 possible numbers from a pool of 'numbered balls'.

You need to fill in the two blanks in the program below so that it works correctly. You cannot alter any of the code given to you or add any new code.

```
import random

print("This is the Mark 6 random number generator.")
print("I will generate 6 random numbers for you.")

balls = []

while len(balls) _____ :
    ball = random.randint(1, 49)


    if balls.count(ball) _____ :
        balls.append(ball)
        print(ball)

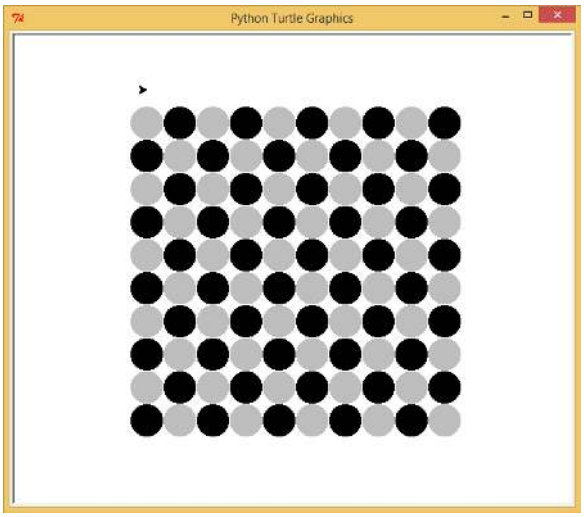
print("Finished! Good luck with these numbers!")
```


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Q3) **12 marks**

A Python program has been written to draw the pattern shown on the right using turtle graphics programming.



The pattern contains gray dots (using the Python colour "gray") and black dots (using the Python colour "black") arranged alternatively in the rows and columns.

The Python program used to draw the pattern is shown below.

You need to fill in the three blanks in the program so that it draws the pattern correctly. You cannot alter any of the code given to you or add any new code.

You may want to know that the `turtle.dot` command can be used to draw a filled circle in the turtle window even if the turtle's pen is up.

```
import turtle

turtle.speed(0)
turtle.up()
turtle.goto(-180, -180)

for y in range(10):
    for x in range(10):
        if x % 2 == y % 2:


            turtle.color( _____ )
        else:


            turtle.color( _____ )
        turtle.dot(40)
        turtle.forward(40)


    turtle.forward( _____ )

    turtle.left(90)
    turtle.forward(40)
    turtle.right(90)

turtle.done()
```


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Q4) **16 marks**

Here is a Python program.

```
import turtle
import random

color_values = ["green", "blue", "red", "purple", "yellow"]

print("This program draws some shapes in random locations.")
print("Please give me two numbers.")

number1 = int(input("First number: "))
number2 = int(input("Second number: "))

turtle.width(5)
turtle.speed(0)

for n in range(number1, number2 + 1):
    length = random.randint(30, 50)

    turtle.up()
    turtle.goto(random.randint(-200, 200), random.randint(-200, 200))
    turtle.down()

    turtle.color(color_values[n % len(color_values)])

    s = (n % 4) + 4
    for _ in range(s):
        turtle.forward(length)
        turtle.right(360 / s)

turtle.done()
```

If you run the program and then enter **18** and **80** as the first number and second number respectively, how many shapes of different colours will be created by the program? Please fill in the number of shapes created in the blanks below.

Answer:

The **total** number of shapes created by the program is: _____

The **total** number of **hexagons** created by the program is: _____

The **total** number of **red squares** created by the program is: _____

The **total** number of **yellow squares** created by the program is: _____

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Q5) **12 marks**

A food store has created the following Python program:

```
food = ["Ice Cream", "Hamburger", "Cake", \
        "Cookies","Chicken Wings","Dim Sum"]

price = [40, 22, 28, 25, 20, 15]

for i in range(len(food)):
    print(i + 1, end=": $")
    print(price[-i-1], end=" - ")
    print(food[i])

item = int(input("Which food item do you want to buy? "))
money = int(input("How much money do you have? "))

pieces = int(money / price[-item])
money = money - pieces * price[-item]

print("You bought", pieces, "piece(s) of", food[item-1])
print("You have", money, "dollar(s) remaining.")
```

Part A) **6 marks**

If you run the program, you will be shown a "menu" in the output of the program. Which one of the following items is among the display inside the menu?

A) `0: $40 - Ice Cream`
B) `1: $15 - Dim Sum`
C) `3: $25 - Cake`
D) `5: $40 - Chicken Wings`
E) `6: $15 - Dim Sum`

Answer (A/B/C/D/E): _____

Part B) **6 marks**

If you run the program and then enter **4** and **300** as the input to the questions respectively, how much money do you have according to the last line of the program output?

Answer:

The amount of money you have according to the last line of the program output is: _____

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Q6) **12 marks**

You have made a guessing game in lab 1. In the game, the player has to guess a secret number generated by the computer. A Python program shown below has been created so that both the player and the computer are competing to guess the secret number as quickly as possible.

```
import random

target = 0
finished = False
lower = 1
upper = 100
count = 0

# Generate the secret number
target = random.randint(1, 100)

while not finished:
    # Handle the computer's guess
    computer_guess = int((lower + upper) / 2)
    if computer_guess > target:
        upper = computer_guess - 1
    elif computer_guess < target:
        lower = computer_guess + 1
    else:
        finished = True

    # Handle the player's guess
    player_guess = int(input("Please enter your guess: "))
    if player_guess < 1 or player_guess > 100:
        print("Please enter an integer number between 1 and 100.")
    elif player_guess > target:
        print("Too high.")
    elif player_guess < target:
        print("Too low.")
    else:
        finished = True

    count = count + 1

    print("My guess is", computer_guess, \
          "and your guess is", player_guess)

# Show the result of the game
if computer_guess != target:
    print("You won! It took you", count, "time(s) to do it!")
elif player_guess != target:
    print("I won! It took me", count, "time(s) to do it!")
else:
    print("Everybody won. It took us", count, "time(s) to do it!")
```


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Part A) **6 marks**

After you run the program, you keep on entering different numbers and here is the output of the program together with your input. The text in bold is your input to the program.

```
Please enter your guess: 70
Too high.
My guess is 50 and your guess is 70
Please enter your guess: 60
Too low.
My guess is 75 and your guess is 60
Please enter your guess: 65
Too low.
My guess is 62 and your guess is 65
Please enter your guess: 68
Too high.
My guess is 68 and your guess is 68
Please enter your guess: 67
My guess is 65 and your guess is 67
```

The last line of the program output is not shown above. Which one of the following is the last line of the program output?

A) `You won! It took you 10 time(s) to do it!`
B) `You won! It took you 5 time(s) to do it!`
C) `I won! It took me 10 time(s) to do it!`
D) `I won! It took me 5 time(s) to do it!`
E) `Everybody won. It took us 5 time(s) to do it!`

Answer (A/B/C/D/E): _____

Part B) **6 marks**

Let's assume that when you run the program the `random.randint` command generates a number **23** as the secret number in the variable `target`.

How many times would it take for the computer to guess the number correctly?

Answer:

The number of times that the computer would take to guess the number correctly is: _____

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Q7) **16 marks**

Peter Pan worked for the CSE Department as a part-time programmer for a total of 240 hours. The initial hourly salary was $56 per hour. His salary was increased by $4 after every 12 hours of work that he had finished. That means the hourly salary was $60 after he had finished a total of 12 hours of work and $64 after a total of 24 hours of work. In addition, he got a special bonus of $56 whenever his hourly salary was increased to a multiple of $5. The hourly salary and bonus he was paid after each hour of work he had finished are shown below:

Number of hours of work finished	Hourly salary	Bonus
1	$56	-
⋮		
12	$56	-
13	$60	$56
⋮		
24	$60	-
25	$64	-
⋮		
72	$76	-
73	$80	$56
⋮		
240	$132	-

To verify whether he was paid correctly, Peter Pan wrote a Python program to calculate the total amount of money he should get for his work. The program is shown below. You need to fill in the six blanks so that the program calculates his salary correctly. You cannot alter any of the code given to you or add any new code.

```
hourly_salary = 56
total_salary = 0

for hour in range(1, 241):


    if _____ != _____ and \


       _____ == _____ :
        hourly_salary = hourly_salary + 4


        if _____ == _____ :
            total_salary = total_salary + 56

    total_salary = total_salary + hourly_salary

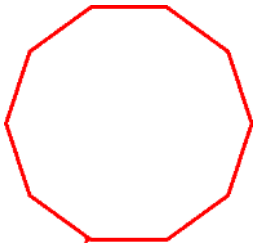
print("My total salary is $", total_salary)
```


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Q8) **6 marks**

After running a Python program, the turtle draws the shape shown on the right inside the turtle window.



The Python program is shown below. Please fill in the three blanks so that the program draws the shape shown above. You cannot alter any of the code given to you or add any new code.

```
import turtle

def myshape( _________________ , _________________ , _________________ ):
    turtle.color(colour)
    for _ in range(sides):
        turtle.forward(length)
        turtle.left(360 / sides)

turtle.width(5)
myshape(10, 100, "red")

turtle.done()
```

We haven't discussed functions yet, so don't worry about this question!

Q9) **12 marks**

This question is related to digital audio but **not** the MIDI music that you have worked on in your labs.

You are given a list of numbers, called `samples`, containing the audio samples from a digital audio file. Each second of audio in this file has 44100 samples. For example, if the length of the given list is 44100, then the digital audio file contains 1 second of audio; if the length of the given list is 88200, then the digital audio file contains 2 seconds of audio.

The digital audio file in `samples` has a duration of **5.5 seconds**. A line of Python code has been used to extract parts of the audio to form a new one:

```
new_samples = samples[:44100*2:2] + samples[int(len(samples)*0.5)::]
```

Please answer the following questions by filling in the duration (in seconds) in the blanks below.

Answer:

What is the duration of audio in `samples[:44100*2:2]`? _____ s

What is the duration of audio in `samples[int(len(samples)*0.5)::]`? _____ s

What is the duration of audio in `new_samples`? _____ s

- End of the exam -

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